EcGFp5 m = 5 quotient: block-Wiedemann finish on the BNS22 chain matrix

Date. 2026-11-02. **Question.** The baseline W0 is the merged quotient solve `research/ecgfp5_quotient_solve_20261101`. Two-thirds of its whole-solve cost is msolve's change of order plus rational parametrisation. Replace that finish with a block-Wiedemann finish on the BNS22 chain matrix of the reduced grevlex basis. Does the whole-solve cost c fall on the same three systems, with every answer verified?

Protocol: `PROTOCOL.md`, frozen and pushed in commit `3d3f68d70` (merged #3297) before the first cell started. There are no amendments.

Status. All three frozen cells completed on one 4-vCPU host (tier L). It is a **measured improvement over the per-system index-calculus baseline (W0) at all three primes**, so it is a pipeline improvement under §5 and becomes the new per-system baseline. It is an *engineering* change: the same F4, a cheaper finish, the same output. It is not a relabelling: every phase is charged, the eliminant is identical, and the planted point is recovered. **The G1 verdict does not change.** The new c_cyc lies below the CLOSE line at every prime, but the frozen [c/4, 4c] band straddles it. This is a per-system stage cost at toy primes. It is not an end-to-end ECDLP cost, and it makes no security claim about EcGFp5.

Every number below is labelled *measured* (from a committed run record), *derived* (a calculation from those records, done in `analyze.py`) or *predicted* (stated in the protocol before the runs).

1. Answer

p	c_new CPU-s	W0 CPU-s	speed-up	bar (W0/1.5)	log2 c_cyc new / W0	G1 band	gates
4111	16,912	31,004	**1.83×**	20,670	45.43 / 46.30	[43.43, 47.43], straddling	all pass
262231	18,823	37,036	**1.97×**	24,690	45.58 / 46.56	[43.58, 47.58], straddling	all pass
33554371	22,516	38,778	**1.72×**	25,852	45.84 / 46.63	[43.84, 47.84], straddling	all pass

c_new is *measured* speed-up, bar and log2 c_cyc are *derived* (c_cyc = CPU-s × 2.8·10^9).

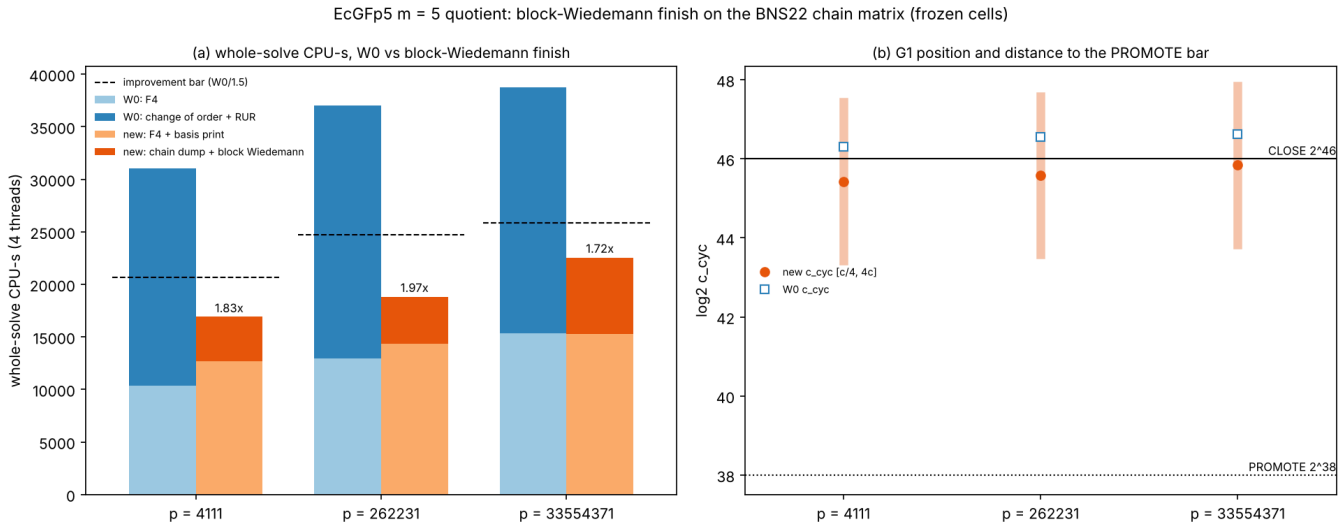
1. **Correct at every prime** (*measured*).
   - The block-Wiedemann eliminant equals W0's msolve eliminant coefficient for coefficient (`cmp` of `bw.elim` and `w0.elim`).
   - Its degree is D = 65,536.
   - The BNS22 stability check passes: 4,920 chains, longest 37, `stab_fail=0`.
   - The planted point is among the recovered rational points: 4, 2 and 2 of them.
2. **Improvement at all three primes.** c_new is 1.72–1.97× below W0, and under the frozen 1.5× bar at every prime. Under §5 this is a pipeline improvement and the new per-system baseline.
3. **The finish alone is 3.1–4.1× cheaper** (*derived*).
   - Here the finish is the extra msolve CPU for printing the basis, plus hnfsolve, plus bwfinish.
   - Against W0's change of order plus parametrisation it is 3.15×, 4.12× and 3.26×.
   - The F4 work is unchanged: it is the same msolve binary on the same input.
4. **G1 verdict unchanged.**
   - The point estimates move below CLOSE (2^46) at every prime: 0.57, 0.42 and 0.16 bits below.
   - The frozen [c/4, 4c] band still crosses the line, so under §5 each cell is reported as *straddling*, not as a verdict change.
   - **Distance remaining to PROMOTE (2^38): 7.43, 7.58 and 7.84 bits** (*derived*). The lead's atlas `revisit_when` (c_cyc ≤ 2^38) is not met.

2. Where the time goes (*measured*, CPU-s)

p	msolve (F4 + basis print)	W0 F4	A/A	hnfsolve	bw sequence	generator	det	points	finish total	finish vs pred
4111	12,655	10,349	+22.3 %	104	3,625	170	73	266	4,153	+6.5 %
262231	14,334	12,981	+10.4 %	134	3,906	194	83	148	4,355	+11.7 %
33554371	15,270	15,325	−0.4 %	126	6,566	198	122	168	7,119	−3.8 %

- **The sequence dominates the finish (87–92 %).** It is L = 1,028 dense GEMMs (65,536 × 4,920)(4,920 × 128) in fp64. At p = 33554371 the operands need two 13-bit limbs, which costs 1.8× the sequence CPU of the one-limb primes, as predicted.
- **Prediction check.** Every finish is within ±30 % of its prediction (−3.8 % to +11.7 %), so there is no prediction failure. Every c_new is within 10 % of its prediction (15,500, 18,100 and 23,900 predicted).
- **A/A control** (msolve CPU against W0's F4 CPU, which differs only by the basis print and host drift).
  - It ranges from −0.4 % to +22.3 %, all inside the frozen ±25 %, so there is no host-drift flag.
  - The spread is wide: printing 306M basis terms should cost about the same at every prime, yet the measured excess ranges from about 0 to 2,300 CPU-s.
  - Read c_new to about ±2,000 CPU-s. Every cell still clears its bar by more than that (by 3,758, 5,867 and 3,336 CPU-s).
- **Memory** (*measured* maxrss).
  - msolve: 4.7–5.2 GiB.
  - bwfinish: 4.85 GiB (one limb) and 9.6 GiB (two limbs).
  - Both are under the 12 GiB envelope. The PML LNZ17 Hermite-form route ran out of memory at 13.3 GiB (`research/ecgfp5_quotient_hnf_finish_20261102`), which is why this finish exists.
- **Wall** (context only): msolve 4,109, 4,515 and 4,809 s. Each cell took 1.5–1.9 h end to end.

3. Figure



Whole-solve CPU-s and G1 position

- **(a)** Whole-solve CPU-s for W0 and the new pipeline, stacked by phase, with each prime's improvement bar.
- **(b)** log2 c_cyc for W0 and the new pipeline, with the frozen [c/4, 4c] band against the CLOSE and PROMOTE lines.

Rendered by `analyze.py` from the run records only.

4. Method (as frozen)

1. msolve 0.6.5 (`-v 2 -t 4 -g 2`) writes the reduced grevlex basis into a FIFO. msolve is the same binary as W0's (sha256 in `results/*/inputs.sha256`).
2. `src/hnfsolve.cpp --dump-P` parses the basis and checks the BNS22 stability assumption. It then writes the tail Q of P = diag(w^{d_k}) + Q (t = 4,920 chains, Σ d_k = D).
3. `src/bwfinish.cpp`:
   - computes the block-Wiedemann sequence S_i = U M_w^i V with b = 128 and seed 1, where M_w = shift − Q^{T}·E_top, so each step is one exact fp64 GEMM through OpenBLAS;
   - computes the minimal generator by PML `pmbasis`, and its determinant;
   - takes the eliminant as the reversal of that determinant;
   - finds the roots by gcd(w^p − w, ·), and each point from ker P(w_0).
4. `run_cell.sh` chains the three steps per prime. `cards/CARD-1.md` is the run card. The three cells ran back to back from 06:17:55Z to 11:20:02Z, with nothing else running on the host.

5. Host, isolation, runs

- **Host.** Tier L: this session's container, 4 vCPU Intel Xeon @ 2.80 GHz, 15.7 GiB. It is the same host class as W0. There is no core pinning. CPU-seconds are getrusage totals.
- **Runs.** Three cells requested, three completed, no failures, no cap stop.
- **Not in Git.** `P.bin` (1.3 GB per prime) is deleted after each cell. Its sha256 is in `results/*/outputs.sha256`, and it is a deterministic function of `quot.ms`. `w0.elim` is regenerated from W0's `quot.out` (artifact store) by `src/elim_from_msolve.py`.
- **Dev runs** in `dev/m5p4111/` (b = 32 and b = 128 on the p = 4111 dump) set the predictions. They are not data.

6. What this does not say

- **Not "EcGFp5 is weaker than claimed".** c is a per-system Gröbner stage cost at toy primes (12–25 bits), not at EcGFp5's 64-bit p, and not an end-to-end ECDLP cost.
- **Not "the lead is reopened".** The atlas asks for c_cyc ≤ 2^38. The new point estimates are 7.4–7.8 bits above it, and the G1 band straddles CLOSE.
- **No leaderboard or `benchmarks/` row.** This stage cost has no matched rho reference (the protocol's scope, §1) and no replay certificate or validator report. Under the validation gate it stays outside the admitted tables. `benchmarks/20261102-ecgfp5-bw-finish/` keeps documentary copies only, as W0's does.
- **No solver lower bound.** F4 is now about two-thirds of c, so the next factor has to come from F4 itself (§7).

7. Next

- **The F4 phase is now the bottleneck** (68–76 % of c). The probes of the merged W0 study (weighted F4, Singular `sba`) gave no improvement. The open levers are:
  - **L3**: the u-coordinate quotient ring A(s) + πB(s), and the variable-order arms (`research/ecgfp5_representation_audit_20261102`);
  - a GVW/F5-style signature F4.
- **Cheaper sequence.** b = 128 is the measured GEMM sweet spot on this host. A modular-BLAS kernel (FFLAS-FFPACK) might cut the two-limb penalty at 25-bit p.
- **Tier-X cell at 64-bit p** once an EKS runner exists. The finish uses multi-limb GEMM there (five 13-bit limbs).

Files

- `PROTOCOL.md`: the frozen protocol.
- `src/`: `hnfsolve.cpp`, `bwfinish.cpp` and `elim_from_msolve.py`, built by `build.sh`. `run_cell.sh` runs one cell.
- `cards/CARD-1.md`: the run card.
- `results/m5_p*/`: the run records (exit codes, msolve run JSON and log, dump log, `bw.json` and `bw.log`, both eliminants, sha256 files, timestamps).
- `dev/`: development runs, not data.
- `analyze.py` produces `analysis.json`, `analysis.md` and `figures/bw_finish.{svg,png}`.
- `build_report.sh` builds `report.pdf`.